

Teacher Guide — senior / module-12- movement-of-sun

IKS Curriculum

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Overview

Movement of the Sun

Band: Senior (Grades 9–12) · **Sessions:** 10 × 45 min · **Sanskrit:** Sūrya-gati — Uttarāyaṇa and Dakṣiṇāyaṇa

Overview

Earth's axial tilt produces an apparent annual path of the Sun along the ecliptic. Indian astronomy names the northward course *Uttarāyaṇa* and southward course *Dakṣiṇāyaṇa*, with 12 *Rāśi* (zodiacal divisions) and *saṁkrānti* moments when the Sun crosses between them. Sāyana (tropical) vs nirayana (sidereal) reference frames are introduced for senior students.

NEP 2020 alignment

- §4.6 — Integration of Indian Knowledge Systems
- §4.23 — Experiential learning (sunrise observation)
- §11.8 — Holistic, multidisciplinary (astronomy + calendar)

Key terms (this module)

- *Sūrya* — the Sun
- *Uttarāyaṇa* — northward solar course (Dec 21 – June 21)
- *Dakṣiṇāyaṇa* — southward solar course (June 21 – Dec 21)
- *saṁkrānti* — Sun's crossing from one Rāśi to the next
- *rāśi* — 1/12th of the ecliptic; zodiacal sign
- *sāyana* — tropical zodiac (linked to seasons)
- *nirayana* — sidereal zodiac (linked to fixed stars)

Prerequisites

Module 2 (Panchabhūta) is recommended before this module. Modules 5 (Bindu Paddhati / dot addition) and 7 (×11 Multiplication) are useful prerequisites for Module 9 and 11.

Materials needed (across 10 days)

- Notebook + pencils per student
- Whiteboard / chart paper
- (See [lesson-plan.md](#) for per-day material lists)

10-day map

- **Day 1** — The Sun's Yearly Path: Observe that sunrise direction changes; introduce the ecliptic.
- **Day 2** — Axial Tilt and the Seasons: Demonstrate with torch+globe why we have summer and winter.
- **Day 3** — Uttarāyaṇa and Dakṣiṇāyana: Identify the dates and direction of each.
- **Day 4** — Equinoxes — Viṣuva: Identify spring and autumn equinoxes (day = night).
- **Day 5** — Twelve Rāśis: Name the 12 rāśis in Sanskrit and pair with Greek zodiac names.
- **Day 6** — Saṁkrānti Moments: Identify Makara, Karka, and other saṁkrāntis on the calendar.
- **Day 7** — Sāyana vs Nirayana: Distinguish tropical and sidereal zodiacs; introduce ayanāṁśa.
- **Day 8** — Precession of the Equinoxes: Explain why Indian and Western zodiac dates differ today.
- **Day 9** — Cross-Cultural Solar Calendars: Compare with Egyptian, Roman, Gregorian.
- **Day 10** — Final Share and Assessment: Project share + summative quiz.

Files in this pack

- [lesson-plan.md](#) — detailed 10-day arc
- [sources.md](#) — sources cited in this module
- [teacher-notes.md](#) — common misconceptions, differentiation
- [parent-guide.md](#) — what parents should know
- [days/day-NN-*.md](#) — 10 per-day lesson plans
- [quizzes/formative.md](#), [quizzes/summative.md](#)
- [activities/activity-01-*.md](#)
- [assessments/rubric.md](#), [assessments/project-brief.md](#)
- [student-workbook.md](#)
- [writing-prompts.md](#) (or [oral-recall.md](#) for Primary)
- [homework.md](#)
- [slides/deck.md](#)

Voice

Analytical, comparative, primary-source aware; 20–30 words on average. Pitches the concept first, frames the tradition as historical context, and uses IAST for all Sanskrit terms.

10-Day Lesson Plan

Lesson Plan — Movement of the Sun (Senior)

10-day arc, 45 min per session.

Day 1 — The Sun's Yearly Path

Objective: Observe that sunrise direction changes; introduce the ecliptic.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 2 — Axial Tilt and the Seasons

Objective: Demonstrate with torch+globe why we have summer and winter.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 3 — Uttarāyaṇa and Dakṣiṇāyaṇa

Objective: Identify the dates and direction of each.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 4 — Equinoxes — Viṣuva

Objective: Identify spring and autumn equinoxes (day = night).

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 5 — Twelve Rāśis

Objective: Name the 12 rāśis in Sanskrit and pair with Greek zodiac names.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 6 — Saṁkrānti Moments

Objective: Identify Makara, Karka, and other saṁkrāntis on the calendar.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 7 — Sāyana vs Nirayana

Objective: Distinguish tropical and sidereal zodiacs; introduce ayanāṁśa.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 8 — Precession of the Equinoxes

Objective: Explain why Indian and Western zodiac dates differ today.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 9 — Cross-Cultural Solar Calendars

Objective: Compare with Egyptian, Roman, Gregorian.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 10 — Final Share and Assessment

Objective: Project share + summative quiz.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Pacing notes

- Senior sessions are 45 min. Adjust if your block is different.
- Days 5 and 9 are buffer days — use to reinforce weaker concepts or extend stronger ones.
- The summative project (see [assessments/project-brief.md](#)) runs in parallel from Day 6 onward.

Differentiation

- **For advanced students:** see [teacher-notes.md](#) for extension prompts.
- **For students needing more support:** simplify activities, do more pair-work, allow oral instead of written responses.

Teacher Notes

Teacher Notes — Movement of the Sun (Senior)

Common student misconceptions

- *“This is religious, not scientific.”* — Frame the tradition as historical observation. Pañcabhūta-like classifications of matter exist in many cultures. Indian astronomy and math are documented, peer-reviewed historical fields.
- *“The Sanskrit terms are just jargon.”* — Insist on the triple format (Sanskrit / IAST / meaning) once, then use the Sanskrit alone. The terms are precise; the discipline of using them properly is itself a learning outcome.
- *“If it’s traditional, it must work.”* — Distinguish empirical claims (testable) from ethical/aesthetic frames (which need separate justification). Encourage students to ask “how would I test this?”.

Differentiation strategies

- **Extension for advanced students:** offer one optional “deeper read” per day — usually a primary source or modern paper.

- **Support for students needing scaffolds:** more pair-work, allow oral responses, provide partial outlines for writing prompts.
- **Multilingual classrooms:** Sanskrit terms are universal; vernacular equivalents (Hindi, Tamil, Telugu, Bengali, Marathi etc.) should be welcomed.

Safety

- Any activity involving food, plants, fire, or outdoor work needs explicit safety briefing.
- See the relevant day file for activity-specific safety notes.

Honest framing

- Where modern science clearly extends or revises traditional claims, say so.
- Where the tradition's framing is best understood as ethical/aesthetic, present it as such (not as competing science).
- Where attribution is debated (e.g., the “Vedic” status of Tirthaji's mathematics), present the debate honestly.

End-of-module reflection (for teachers)

- Which concepts landed well? Which didn't?
- Which students grew most? Which need a different approach next module?
- Add notes to `teacher-notes.md` for the next teacher who uses this pack.

Sources

Sources — Movement of the Sun (Senior)

Primary and secondary sources

- Sūrya-siddhānta (c. 5th–10th c. CE), chapters on solar motion
- Āryabhaṭīya (499 CE), on precession
- Varāhamihira, *Br̥hat Saṁhitā* (6th c. CE)
- NCERT Physics Class XI, ch. 8 (Gravitation)
- NCERT Geography Class IX, ch. 3 (Motions of the Earth)

A note on citation discipline

Per the curriculum's style guide, citations must be verifiable. If a verse number, page, or attribution cannot be confirmed, the source is paraphrased without invented attribution. Where direct quotes appear in Senior-band day files, the source is identified in IAST + edition.

For deeper reading

- The shared glossary at `curriculum/_shared/glossary.md` contains the IAST + meaning for every Sanskrit term used.

- Other modules in this curriculum that connect: see the module README’s “Prerequisites” section.

Day 1 — The Sun’s Yearly Path

Module: Movement of the Sun · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: observe that sunrise direction changes; introduce the ecliptic.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *Sūrya* (IAST: Sūrya; lit. “the Sun”)

Primary source (today’s reading)

From *Sūrya-siddhānta* (c. 5th–10th c. CE) — see `sources.md`.

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **The observation.** Project a year-long animation of sunrise direction (or show a static composite of 12 sunrise photos taken from the same spot, one per month). “*The Sun rises in roughly the east — but the EXACT direction shifts north and south through the year.*”
2. **Define the ecliptic.** The Sun’s apparent path against the background stars over one year. A great circle on the celestial sphere, tilted $\sim 23.5^\circ$ from the celestial equator.
3. **Why does it shift?** Earth’s rotational axis is tilted $\sim 23.5^\circ$ relative to its orbital plane. As Earth orbits the Sun, this tilt projects onto our sky as the Sun rising/setting north of due-east in summer (NH), south in winter.
4. **Source.** *Sūrya-siddhānta* ch. 2 — “*sūryasya gatiḥ*” — gives a mean and true solar motion model. Specifies daily and annual displacement on the ecliptic.

5. **Independent task.** Each student looks up their city’s sunrise direction today vs 6 months from today (web). Note the difference in compass bearing.

Activity (15 min)

A scaled version of *Sunrise Direction Tracking* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces The Sun’s Yearly Path.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 2 — Axial Tilt and the Seasons

Module: Movement of the Sun · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: demonstrate with torch+globe why we have summer and winter.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See [lesson-plan.md](#) for any day-specific materials)

Vocabulary introduced today

- *Uttarāyaṇa* (IAST: *Uttarāyaṇa*; lit. “northward solar course (Dec 21 – June 21)”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Torch + globe demo.** Hold a globe with the axis tilted $\sim 23.5^\circ$ (mark the North Pole with tape). Shine a strong torch from one direction (the Sun). Slowly rotate the globe and observe which hemispheres get more direct light.
2. **Key observation.** When the Northern Hemisphere tilts TOWARD the Sun (June solstice), it gets more direct sunlight, longer days. Southern Hemisphere experiences the opposite (winter).
3. **Source.** Sūrya-siddhānta and Āryabhaṭīya both note Earth’s globe-shape (*gola*) and the tilt. Āryabhaṭa correctly identifies the rotation cause of day-night ~ 499 CE.
4. **Modern bridge.** NCERT Physics XI ch. 8 gives the modern derivation: Earth’s axial tilt + orbital geometry $\rightarrow$ annual solar declination $\rightarrow$ seasons.
5. **Discussion.** “*Why are January and July temperatures so different — even though Earth is at almost the same distance from the Sun?*” (Answer: tilt, not distance.)

Activity (15 min)

A scaled version of *Sunrise Direction Tracking* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Axial Tilt and the Seasons.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 3 — Uttarāyaṇa and Dakṣiṇāyana

Module: Movement of the Sun · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: identify the dates and direction of each.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *Dakṣiṇāyana* (IAST: Dakṣiṇāyana; lit. “southward solar course (June 21 – Dec 21)”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Define the two ayanas.**
 - *Uttarāyaṇa* — northward course. Roughly Dec 21 (winter solstice) through Jun 21 (summer solstice). Sun moves NORTH on the sky (Northern Hemisphere).
 - *Dakṣiṇāyana* — southward course. Jun 21 through Dec 21. Sun moves SOUTH.
2. **Cultural importance.** *Uttarāyaṇa* is considered auspicious in many Indian traditions. *Makara Saṅkrānti* (~Jan 14) is celebrated as the start of *Uttarāyaṇa* — but technically the solstice (Dec 21) IS the start. The Jan 14 date reflects the sidereal-zodiac (*nirayana*) convention. We’ll unpack this on Day 7.
3. **Source.** Sūrya-siddhānta defines both ayanas in chapter 2.
4. **Modern bridge.** The Northern Hemisphere’s summer happens during the Sun’s *Uttarāyaṇa* (high northern declination). NCERT Geography IX ch. 3 has the standard diagram.
5. **Identify today.** “*Which ayana are we in today?*” Students compute based on the date. (Northern winter to summer = *Uttarāyaṇa*; rest = *Dakṣiṇāyana*.)

Activity (15 min)

A scaled version of *Sunrise Direction Tracking* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces *Uttarāyaṇa* and *Dakṣiṇāyana*.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See homework.md for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 pañcabhūta).

Day 4 — Equinoxes — Viṣuva

Module: Movement of the Sun · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: identify spring and autumn equinoxes (day = night).

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *saṁkrānti* (IAST: saṁkrānti; lit. “Sun’s crossing from one Rāśi to the next”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Define equinox.** *Viṣuva* in Sanskrit. The two days each year (Mar 21 ± 1 and Sep 22 ± 1) when day length equals night length everywhere on Earth.

2. **Why?** At equinox, Earth's axis is perpendicular to the Sun-Earth line. Sunlight falls equally on both hemispheres. The Sun rises due-east and sets due-west.
3. **Source.** Sūrya-siddhānta names the *viṣuva* points. Varāhamihira (*Bṛhat Saṃhitā*) discusses them in calendar context. The Vernal Equinox is the traditional anchor of the *sāyana* (tropical) zodiac.
4. **Cross-cultural.** The Roman / Gregorian calendar implicitly aligns to the equinoxes. Easter is computed by lunar-solar reckoning anchored to the Vernal Equinox. Many cultures celebrate the equinoxes — Holi (~Phālguna pūrṇimā, near Vernal Equinox), Mabon (NH autumn).
5. **Observation activity.** Each student notes the sunrise direction this week (or projects a stellarium app). Compare to expected (near-due-east if we're near an equinox).

Activity (15 min)

A scaled version of *Sunrise Direction Tracking* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces Equinoxes — Viṣuva.

Wrap-up (5 min)

- One-sentence exit ticket: “*What's one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today's task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 5 — Twelve Rāśis

Module: Movement of the Sun · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: name the 12 rāśis in sanskrit and pair with greek zodiac names.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *rāśi* (IAST: *rāśi*; lit. “1/12th of the ecliptic; zodiacal sign”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Define *rāśi*.** 1/12th of the ecliptic = 30° each. The Sun spends roughly one month in each *rāśi*.
2. **The 12 Sanskrit names.** *Meṣa*, *Vṛṣabha*, *Mithuna*, *Karka*, *Siṃha*, *Kanyā*, *Tulā*, *Vṛścika*, *Dhanus*, *Makara*, *Kumbha*, *Mīna*.
3. **Map to Greek/Western zodiac.** *Meṣa* = Aries, *Vṛṣabha* = Taurus, ... *Mīna* = Pisces. The 12-fold division of the ecliptic is widespread across cultures (Babylonian, Greek, Indian, possibly via cultural contact).
4. **Source.** *Br̥hat Saṃhitā* of Varāhamihira (6th c. CE) systematises the 12 *rāśis*. Earlier references in *Garga Saṃhitā*.
5. **Map current month to current *rāśi*.** Looking at a sidereal pañcāṅga, identify which *rāśi* the Sun is in this month. Note the difference from Western (tropical) zodiac dates — we’ll resolve this on Day 7.

Activity (15 min)

A scaled version of *Sunrise Direction Tracking* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Twelve Rāśis.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 6 — Saṅkrānti Moments

Module: Movement of the Sun · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: identify makara, karka, and other saṅkrāntis on the calendar.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *sāyana* (IAST: *sāyana*; lit. “tropical zodiac (linked to seasons)”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Define saṅkrānti.** The moment the Sun moves from one *rāśi* to the next. ~12 saṅkrāntis per year (one every ~30 days). Two are most culturally prominent: *Makara Saṅkrānti* (Capricorn entry) ~Jan 14, and *Karka Saṅkrānti* (Cancer entry) ~Jul 16.
2. **The Jan 14 / Solstice mystery.** *Makara Saṅkrānti* is celebrated as the start of *Uttarāyaṇa* — but Dec 21 (solstice) is the astronomical start. Why the ~24-day difference? Answer (preview): the celebration uses the *SIDEREAL* (nirayana) zodiac, while the seasons (and solstice) align with the *TROPICAL* (*sāyana*) zodiac. Tomorrow we’ll unpack this.

3. **Source.** Sūrya-siddhānta computes saṅkrāntis as solar longitudes crossing rāśi boundaries.
4. **Cultural saṅkrāntis.** Festivals tied to specific saṅkrāntis: *Makara Saṅkrānti* (Pongal in Tamil Nadu, Lohri in Punjab, Magh Bihu in Assam), *Karka Saṅkrānti* (mid-monsoon).
5. **Class exercise.** Identify the 12 saṅkrānti dates for the current year (use a published pañcāṅga). Plot them on a circular year-wheel.

Activity (15 min)

A scaled version of *Sunrise Direction Tracking* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces Saṅkrānti Moments.

Wrap-up (5 min)

- One-sentence exit ticket: “What’s one thing you learned today?”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 pañcabhūta).

Day 7 — Sāyana vs Nirayana

Module: Movement of the Sun · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: distinguish tropical and sidereal zodiacs; introduce ayanāṁśa.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See `lesson-plan.md` for any day-specific materials)

Vocabulary introduced today

- *nirayana* (IAST: nirayana; lit. “sidereal zodiac (linked to fixed stars)”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Set up the conflict.** Indian astronomy distinguishes two reference frames:
 - *Sāyana* (with-ayana, “tropical”) — measured from the Vernal Equinox.
 - *Nirayana* (without-ayana, “sidereal”) — measured from a fixed point in the sky (a star, typically near Citrā/Spica).
2. **The drift.** Earth’s axis wobbles (precession of equinoxes), causing the Vernal Equinox point to shift slowly westward against the fixed stars at ~50 arcseconds per year. Over ~2000 years, the cumulative drift is ~28°. *That* is the ~24° gap between Dec 21 (solstice in *sāyana*) and Jan 14 (when the Sun enters Makara in *nirayana*).
3. **Source.** Āryabhaṭīya documents the precession concept; Bhāskara II refines it. Indian *jyotiṣa* preserves the *nirayana* convention; modern astronomy uses *sāyana*.
4. **Why does this matter?** Festival dates anchored to *nirayana* drift slowly relative to the seasons. Over thousands of years, *Makara Saṅkrānti* will shift even further from the December solstice unless re-anchored.
5. **Critical thinking.** “*Should jyotiṣa update its zodiac to sāyana? Or preserve the nirayana convention for cultural continuity?*” Both views have advocates. There’s no single right answer.

Activity (15 min)

A scaled version of *Sunrise Direction Tracking* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces *Sāyana* vs *Nirayana*.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.

- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 pañcabhūta).

Day 8 — Precession of the Equinoxes

Module: Movement of the Sun · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: explain why indian and western zodiac dates differ today.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *Sūrya* (IAST: Sūrya; lit. “the Sun”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **The phenomenon.** Earth’s axis wobbles like a slow top, completing one full circle every ~25,800 years. Rate: ~50.3 arcseconds per year. Discovered by Hipparchus ~130 BCE.
2. **Indian computation.** Āryabhaṭīya (499 CE) implicitly includes precession in its longitude calculations. Later Indian astronomers (Bhāskara II, Nīlakaṇṭha) refine the value.
3. **Modern value.** ~50.27”/year — very close to traditional Indian estimates.
4. **Consequence for the zodiac.** The Vernal Equinox point shifts westward through the sidereal zodiac over time. ~2000 years ago, the Vernal Equinox was in Aries (which is why Aries is the “first” zodiac sign in Western astrology). Today, the Vernal Equinox is in Pisces.

5. **Modern source.** NASA / IAU references. Discussion: “*How long until the Sun is in Aquarius at the Vernal Equinox?*” (~600 years.)

Activity (15 min)

A scaled version of *Sunrise Direction Tracking* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Precession of the Equinoxes.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 9 — Cross-Cultural Solar Calendars

Module: Movement of the Sun · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: compare with egyptian, roman, gregorian.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See [lesson-plan.md](#) for any day-specific materials)

Vocabulary introduced today

- *Uttarāyaṇa* (IAST: *Uttarāyaṇa*; lit. “northward solar course (Dec 21 – June 21)”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **The Egyptian solar calendar.** 365 days, 12 months of 30 days + 5 epagomenal days. No leap year initially. Synchronised with the Nile flood.
2. **The Roman / Julian / Gregorian.** Julian (45 BCE) — 365.25 days, leap year every 4. Gregorian (1582) — refined to 365.2425, leap years every 4 except centuries unless divisible by 400.
3. **The Indian solar calendar.** *Saura māsa* = time Sun spends in one rāśi. ~30-31 days per month (varies because Sun moves faster near perihelion). Used in Tamil Nadu, Kerala, Bengal — these regions use sidereal solar months even when their festivals are lunar.
4. **The Chinese solar calendar.** 24 *jiéqì* (solar terms) anchored to specific solar longitudes. Spring Equinox, Summer Solstice, etc. are explicit anchor points.
5. **Comparison exercise.** Each student picks 2 traditions and draws a side-by-side year-circle, noting where their respective “new year” falls and what it celebrates.

Activity (15 min)

A scaled version of *Sunrise Direction Tracking* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Cross-Cultural Solar Calendars.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 10 — Final Share and Assessment

Module: Movement of the Sun · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: project share + summative quiz.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *Dakṣiṇāyana* (IAST: Dakṣiṇāyana; lit. “southward solar course (June 21 – Dec 21)”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Personal solar calendar share (10 min).** Each student shows their year-wheel with 2 solstices, 2 equinoxes, 12 saṅkrāntis, and their own birthday’s rāśi marked.
2. **Vocabulary lightning round (5 min).** Definitions taken from random students: *uttarāyaṇa*, *dakṣiṇāyana*, *viṣuva*, *rāśi*, *saṅkrānti*, *sāyana*, *nirayana*, *ayanāmśa*.
3. **Summative quiz (25 min).** See quizzes/summative.md.
4. **Critical-thinking close.** “*Is the difference between sāyana and nirayana a ‘wrong vs right’ issue or a ‘context-dependent convention’ issue? Which framing is more scientifically honest?*”
5. **Safety reminder.** Never look at the Sun directly. Reaffirm.

Activity (15 min)

A scaled version of *Sunrise Direction Tracking* (see activities/). Today’s slice: focus on the part of the activity that reinforces Final Share and Assessment.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See homework.md for today's task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 pañcabhūta).

Formative Quiz — Movement of the Sun (Senior)

When: mid-module (after Day 5). **Duration:** 25 minutes. **Format:** mix of short answer, multiple choice, and reflection.

Questions

1. Name the three / four key terms introduced in this module so far. Give the IAST and one-line meaning of each.
2. State the central idea of Movement of the Sun in one sentence.
3. (Multiple choice) Which day in this module focused on 'Uttarāyaṇa and Dakṣiṇāyana'?
(a) Day 1 (b) Day 2 (c) Day 3 (d) Day 4
4. True / False — and explain in one sentence: *Movement of the Sun is purely a religious concept and has no scientific framing.*
5. Connect Movement of the Sun to **one** other module in this curriculum. Which module, and how?
6. List **two** sources cited so far in this module.
7. (Short answer) Why does the module distinguish 'Sūrya' from related concepts? Give one example.
8. (Short answer) In your own words, explain Day 3's concept.
9. Identify **one** common misconception about Movement of the Sun and rebut it in one sentence.
10. Write **one** question you have about this module so far — something you want clarified before the summative.

How this is used

This quiz is **formative** — it's a check-in to see what's landing, not a grade. Teacher reviews answers to adjust pacing for Days 6–10. Students may swap and self-mark.

Summative Quiz — Movement of the Sun (Senior)

When: end of module (after Day 10). **Duration:** 45 minutes. **Total points:** 30.

Questions

Part A — Vocabulary (~5 points) A1. Define *Sūrya* in your own words. A2. Define *Uttarāyaṇa* in your own words. A3. Define *Dakṣiṇāyana* in your own words. A4. Define *saṃkrānti* in your own words. A5. Define *rāśi* in your own words.

Part B — Concepts (~10 points) B1. Explain the central idea of Movement of the Sun in your own words (3–4 sentences). B2. Describe the method or framework students used in Day 4 of this module. B3. Identify ONE source cited in this module and explain why it matters. B4. Compare Movement of the Sun with a concept from another module (your choice). What's similar, what's different? B5. Identify ONE common misconception about this module and rebut it in 2–3 sentences.

Part C — Application (~5 points) C1. Give an example of Movement of the Sun from your own daily life. C2. Describe one way this module connects to a current real-world issue (or scientific question).

Part D — Reflection (~5 points; ungraded for content) D1. What surprised you most in this module? D2. What's one question you still have?

Part E — Primary source (~5 points; Senior only) E1. Quote or paraphrase one passage from a primary source cited in this module. Explain it in your own words.

Marking guidance

- Vocabulary: give credit for correct meaning, even if exact wording differs.
- Concepts: look for evidence of understanding, not memorisation.
- Application: any reasonable example accepted.
- Reflection: ungraded for content; awarded for thoughtfulness.

Pass mark

A score of 18/30 indicates the student has met the module's learning objectives.

Homework

Homework — Movement of the Sun (Senior)

Light daily tasks. Each takes 10–15 minutes.

1. **Day 1 — The Sun's Yearly Path** · Write 2 sentences about today's concept. (~10 min)
2. **Day 2 — Axial Tilt and the Seasons** · Write 2 sentences about today's concept. (~10 min)

3. **Day 3 — Uttarāyaṇa and Dakṣiṇāyana** · Write 2 sentences about today's concept. (~10 min)
4. **Day 4 — Equinoxes — Viṣuva** · Write 2 sentences about today's concept. (~10 min)
5. **Day 5 — Twelve Rāśis** · Write 2 sentences about today's concept. (~10 min)
6. **Day 6 — Saṁkrānti Moments** · Write 2 sentences about today's concept. (~10 min)
7. **Day 7 — Sāyana vs Nirayana** · Write 2 sentences about today's concept. (~10 min)
8. **Day 8 — Precession of the Equinoxes** · Write 2 sentences about today's concept. (~10 min)
9. **Day 9 — Cross-Cultural Solar Calendars** · Write 2 sentences about today's concept. (~10 min)
10. **Day 10 — Final Share and Assessment** · Write 2 sentences about today's concept. (~10 min)

Notes

- Homework is intended to consolidate what was learned in class, not introduce new content.
- If a child struggles, the parent should consult the teacher rather than push.
- Capstone project days (Module 15) may have lighter daily homework so students can focus on the project.

Writing Prompts

Writing Prompts — Movement of the Sun (Senior)

Use these for in-class writing, homework, or assessment. Each prompt is suitable for a 300-word response.

1. In 3–4 sentences, explain Movement of the Sun to a friend who hasn't taken this module.
2. Choose one Sanskrit term from this module. Define it, then describe one observation from your own life that matches.
3. Compare Movement of the Sun with a similar concept from another module or from your science textbook. What's similar? Different?
4. Identify ONE claim in this module that you'd want to test. How would you test it?
5. What was the most surprising thing in this module? Why did it surprise you?
6. Pick one source cited in this module. Write 2 sentences explaining why it matters.
7. What is one common misconception about Movement of the Sun? Rebut it in 2–3 sentences.
8. Imagine you are teaching this module to a younger student. What would you say first?
9. Read one primary-source passage cited in this module. Translate or paraphrase it; comment on what it adds to your understanding.
10. Identify a contemporary debate about the IKS framework presented here (e.g., is Tirthaji's mathematics genuinely Vedic? are doṣas a valid framework today?). Take a position with two pieces of evidence.

How to use

- Pick **one** prompt per day for the writing-prompts homework.
- Or assign all of them across the module, alternating between in-class and home.
- Senior students may treat any prompt as a 500-word essay topic.

Activity 1 — Sunrise Direction Tracking

Module: Movement of the Sun · **Band:** Senior · **Duration:** 30 min

Purpose

Reinforce the day-by-day concepts of Movement of the Sun through a hands-on activity. Aligned with NEP 2020 §4.23 (experiential learning).

Materials

compass app, classroom window, notebook

What students do

Over 5 days, students note the compass bearing of sunrise each morning (from the same window). They graph the bearing and predict how it will change in 3 months.

Step-by-step

1. **Setup (5 min)** — gather materials; explain the task.
2. **Working phase (20 min)** — students work individually or in pairs.
3. **Share (5 min)** — selected students share their results with the class.

Differentiation

- **Extension:** ask advanced students to do an additional iteration or analysis.
- **Support:** pair students who need scaffolds; offer partial templates.

Assessment

This activity is **formative**. The teacher observes participation, accuracy, and reflection quality. No grade is assigned.

Safety

- General safety briefing before any materials are handled.
- If herbs, plants, or food are involved: kitchen-amount only, no ingestion of unknown preparations.
- Outdoor activities: stay in the designated area; supervised by the teacher.

Project Brief

Project Brief — Personal Solar Calendar

Module: Movement of the Sun · **Band:** Senior

What you will do

Make a one-page solar calendar for the current year: mark the two solstices, two equinoxes, all 12 saṃkrānti dates, plus your own birthday with its rāśi.

Why

This project asks you to apply concepts from this module to your own life or a real situation. It is graded on the rubric in `rubric.md`.

Deliverables

- **One-page project plan** (Day 2 of the module): problem, sources, deliverable, timeline.
- **Daily journal entries** (one per day, starting Day 6): what you did, what worked, what didn't.
- **Final artefact** (a poster / model / digital deliverable).
- **Written reflection** (500 words (Senior)): what you learned, what surprised you, what you're still uncertain about.
- **Presentation** (5-minute share with the class).

Timeline

- **Day 1–2:** scope and plan.
- **Day 3–4:** research.
- **Day 5:** mid-project critique.
- **Day 6–7:** build / make.
- **Day 8:** document.
- **Day 9:** rehearse.
- **Day 10:** share day.

Sources

You must cite at least 3 sources. Use the form: *Author, Title, year*. (See `../sources.md` for examples.)

Safety

- Any project involving plants / food / outdoor work must be supervised.
- **No herbal medicine recommendations.** Modules 10 and 4 give awareness; therapeutic decisions belong to qualified practitioners.

Grading

See `rubric.md`. The project is worth the same weight as the summative quiz.

Rubric

Assessment Rubric — Movement of the Sun (Senior)

This rubric applies to the project (see `project-brief.md`) and may also inform the summative quiz.

Criteria

Accuracy of concepts — uses Sanskrit terms correctly, gets the central idea right

- **Exceeding (4)** — Exceptional; could teach this to another student.
- **Meeting (3)** — Solid; meets the learning objective.
- **Approaching (2)** — Partial; some elements present, others missing.
- **Beginning (1)** — Significant gaps; needs re-teaching.

Source discipline — cites sources, distinguishes traditional framing from empirical claim

- **Exceeding (4)** — Exceptional; could teach this to another student.
- **Meeting (3)** — Solid; meets the learning objective.
- **Approaching (2)** — Partial; some elements present, others missing.
- **Beginning (1)** — Significant gaps; needs re-teaching.

Application — connects the module to other modules and to real-world examples

- **Exceeding (4)** — Exceptional; could teach this to another student.
- **Meeting (3)** — Solid; meets the learning objective.
- **Approaching (2)** — Partial; some elements present, others missing.
- **Beginning (1)** — Significant gaps; needs re-teaching.

Presentation — clear, organised, evidence-supported

- **Exceeding (4)** — Exceptional; could teach this to another student.
- **Meeting (3)** — Solid; meets the learning objective.
- **Approaching (2)** — Partial; some elements present, others missing.
- **Beginning (1)** — Significant gaps; needs re-teaching.

Reflection — honest, thoughtful, identifies what was learned and what remains uncertain

- **Exceeding (4)** — Exceptional; could teach this to another student.
- **Meeting (3)** — Solid; meets the learning objective.

- **Approaching (2)** — Partial; some elements present, others missing.
- **Beginning (1)** — Significant gaps; needs re-teaching.

How the teacher uses this rubric

1. After the project share, score each student on each criterion.
2. Convert to a band (4 = Exceeding, 3 = Meeting, etc.) for the report card.
3. Share at least one strength and one area for growth with each student.

Self-assessment (Senior students)

Senior students complete a parallel self-assessment using the same rubric before the teacher's assessment. The teacher and student then meet to compare.